

236. Lowest Common Ancestor of Binary Tree

PLATFORM

Platform: LeetCode

DIFFICULTY

MEDIUM

DATE

Solved: July 30, 2026

Problem Summary

Given the root of a binary tree and two nodes p and q, return their lowest common ancestor — the deepest node that has both p and q as descendants (a node can be a descendant of itself).

Approach

Post-order recursion. Search both subtrees for p and q. If both subtrees return non-null, current node is the LCA. Otherwise propagate the non-null result upward.

Algo

1. If root == null return null.
2. If root == p or root == q return root.
3. left = LCA(root.left, p, q).
4. right = LCA(root.right, p, q).
5. If both non-null → return root (LCA found).
6. Return left != null ? left : right.

Observations

- For BST specifically, LCA can be found more efficiently using BST property (LC 235) — no need to search both subtrees.
- The function returns a TreeNode not a boolean — it carries both "found" signal and the actual node reference.
- left != null ? left : right elegantly handles three cases: only left found, only right found, neither found (returns null).

Time Complexity & Space Complexity

$O(n)$ — visits every node in worst case.

$O(h)$ — recursion stack.

Key Insights

- If a node equals p or q, return it immediately — no need to search deeper (problem guarantees both exist in tree).
- Both subtrees returning non-null means p and q are on **opposite sides** — current node is their LCA.
- If only one side returns non-null, both p and q are in that subtree — propagate that result up.
- Works for the case where one node is ancestor of the other — caught by the root == p || root == q check.

Problem List

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Solved

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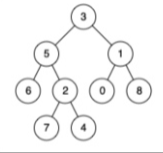
Topics

Companies

Given a binary tree, find the lowest common ancestor (LCA) of two given nodes in the tree.

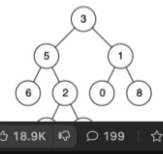
According to the definition of LCA on Wikipedia: "The lowest common ancestor is defined between two nodes p and q as the lowest node in T that has both p and q as descendants (where we allow a node to be a descendant of itself)."

Example 1:



Input: root = [3,5,1,6,2,0,8,null,null,7,4], p = 5, q = 1
Output: 3
Explanation: The LCA of nodes 5 and 1 is 3.

Example 2:



Accepted

33 / 33 testcases passed

Sahil Khan submitted at Jul 30, 2026 18:12

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Runtime

14 ms Beats 89.10%

Memory

69.84 MB Beats 54.99%

Code | Java

```
1 /**
2  * Definition for a binary tree node.
3  * public class TreeNode {
4  *     int val;
5  *     TreeNode left;
6  *     TreeNode right;
7  *     TreeNode(int x) { val = x; }
8  * }
9  */
10 class Solution {
11     public TreeNode lowestCommonAncestor(TreeNode root, TreeNode p, TreeNode q) {
12         if(root == null) return null;
13         if(p==root || q==root) return root;
14
15         TreeNode left = lowestCommonAncestor(root.left, p, q);
16         TreeNode right = lowestCommonAncestor(root.right, p, q);
17
18         if(left!=null && right!=null) return root;
19         return (left!=null) ? left : right;
20     }
21 }
```

Testcase | Test Result

Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

root = [3,5,1,6,2,0,8,null,null,7,4]